

Theorem. $X_1, \dots, X_N$ - i.i.d. $y: X_1, y$ are dependent and y and $X_2, \dots, X_N$ are indep.

$$I(X_1; y) \geq -\mathcal{L}_{NCE}$$

$$\mathcal{L}_{NCE} = \mathbb{E}_{p(x_{1:N}, y)} \left[-\log \frac{e^{f(x_1; y)}}{\frac{1}{N} \sum_{n=1}^N e^{f(x_n; y)}} \right]$$

$$1) I(X_1; y) = KL(p(x_1, y) \parallel p(x_1)p(y)) =$$

$$= \int p(x_1, y) \log \frac{p(x_1, y)}{p(x_1)p(y)} dx_1 dy =$$

$$= \int p(x_1, y) \log \frac{p(x_1|y)}{p(x_1)} dx_1 dy =$$

$$= \int p(x_1, y) \log \frac{p(x_1|y) q(x_1|y)}{p(x_1) q(x_1|y)} dx_1 dy =$$

$$= - \int p(x_1, y) \log p(x_1) dx_1 dy + \mathbb{E}_{p(x_1, y)} \log q(x_1|y) +$$

$$+ \int p(x_1|y)p(y) \log \frac{p(x_1|y)}{q(x_1|y)} dx_1 dy =$$

$$= - \int \left(\int p(x_1, y) dy \right) \log p(x_1) dx_1 +$$

$$+ \mathbb{E}_{p(x_1, y)} \log q(x_1, y) + \underbrace{\mathbb{E}_{p(y)} \text{KL}(p(x_1, y) \| q(x_1, y))}_{\geq 0}$$

$$\geq H(x_1) + \mathbb{E}_{p(x_1, y)} \log q(x_1, y)$$

$$2) \quad q(x_1, y) = \frac{1}{z(y)} p(x_1) e^{g(x_1, y)} \quad z(y) = \int p(x_1) e^{g(x_1, y)} dx_1$$

$g(x_1, y)$ - arbitr.

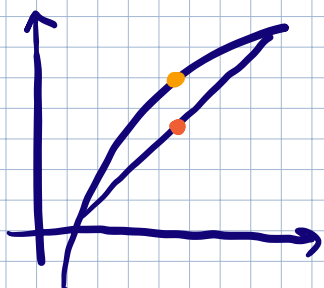
$$\mathbb{E}_{p(x_1, y)} [\log q(x_1, y)] = -\mathbb{E}_{p(y)} \log z(y) + \underbrace{\mathbb{E}_{p(x_1)} \log p(x_1)}_{-H(x_1)} +$$

$$+ \mathbb{E}_{p(x_1, y)} g(x_1, y)$$

$$I(x_1; y) \geq \mathbb{E}_{p(x_1, y)} g(x_1, y) - \mathbb{E}_{p(y)} \log z(y)$$

$$I(x_1, y) \geq \mathbb{E}_{p(x_1, y)} g(x_1, \dots, x_n, y) - \mathbb{E}_{p(y)} \log z(y, x_2, \dots, x_n)$$

$$I(x_1; y) \geq \mathbb{E}_{p(x_1, y) p(x_2:n)} g(x_{1:n}, y) - \mathbb{E}_{p(y) p(x_2:n)} \log z(y, x_2:n)$$



$$\log \mathbb{E} \xi \geq \mathbb{E} \log \xi$$

$$I(x_1; y) \geq \mathbb{E}_{p(x_1, y) p(x_{2:n})} g(x_{1:n}, y) - \log \mathbb{E}_{p(y) p(x_{2:n})} z(y, x_{2:n})$$

$$g(x_{1:n}, y) = \log \frac{e^{f(x_1, y)}}{\sum_{n=1}^N e^{f(x_n, y)}}$$

$$I(x_1; y) \geq \underbrace{\mathbb{E}_{p(x_{1:n}, y)} \log \frac{e^{f(x_1, y)}}{\sum_{n=1}^N e^{f(x_n, y)}}}_{-h_{X|Y} - \log N} - \underbrace{\log \mathbb{E}_{p(y) p(x_{2:n})} z(y, x_{2:n})}_{= \log N}$$

$$\mathbb{E}_{p(y) p(x_{2:n})} z(y, x_{2:n}) = \mathbb{E}_{p(y) p(x_1) p(x_{2:n})} \frac{e^{f(x_1, y)}}{\sum_{n=1}^N e^{f(x_n, y)}}$$

$$= \mathbb{E}_{p(y)} \mathbb{E}_{p(x_{1:n})} \frac{e^{f(x_1, y)}}{\sum_{n=1}^N e^{f(x_n, y)}} =$$

$$= \frac{1}{N} \mathbb{E}_{p(y)} \sum_{i=1}^N \mathbb{E}_{p(x_{1:n})} \frac{e^{f(x_i, y)}}{\sum_{n=1}^N e^{f(x_n, y)}} =$$

$$= \frac{1}{N} \mathbb{E}_{p(y)} \mathbb{E}_{p(x_{1:n})} \frac{\sum_{i=1}^N e^{f(x_i, y)}}{\sum_{n=1}^N e^{f(x_n, y)}} = \frac{1}{N}$$

$$\begin{aligned}
 h_{NCE} &= -\mathbb{E}_{p(x_{1:N}, y)} \left[\log \frac{e^{f(x_1; y)}}{\frac{1}{N} \sum_{n=1}^N e^{f(x_n; y)}} \right] = \\
 &= -\mathbb{E}_{p(x_{1:N}, y)} \left[\log \frac{e^{f(x_1; y)}}{\sum_{n=1}^N e^{f(x_n; y)}} \right] - \log N \\
 &= h_{NCE} + \log N
 \end{aligned}$$

$$\mathbb{E}_{p(y)} KL(p(x_1|y) \| q(x_1|y)) \geq 0$$

$$\begin{aligned}
 q(x_1|y) &\propto p(x_1) e^{g(x_1, y)} \propto p(x_1) \cdot \frac{e^{f(x_1; y)}}{\sum_{n=1}^N e^{f(x_n; y)}} \\
 &\parallel \\
 p(x_1|y) & \\
 \frac{e^{f(x_1; y)}}{\sum_{n=1}^N e^{f(x_n; y)}} &\approx \frac{p(x_1|y)}{p(x_1)}
 \end{aligned}$$